

Network Topologies

Module 15

Duration: 27:15

Video: <https://www.youtube.com/watch?v=wwabRIG7MvU>

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my students hello I welcome you all in our series of lectures on cyber security today we are going to study about computer network basics in this lecture we will learn about various Network types and topologies we will also learn how to identify basic topologies and their variations and how to determine appropriate topology for a given Network plan so let us begin with it a network is a set of devices often referred to as nodes connected by communication links a node can be a computer printer or any other device capable of sending and receiving data generated by other nodes on the network most networks use distributed processing in which a task is divided among multiple computers instead of one single large machine being responsible for all

aspects of a process separate computers usually personal computers or workstations handle a subject a network must be able to meet a certain number of criteria the most important of these are number one performance consistency reliability recovery and security let us study all the criteria one by one number one performance performance is defined as the rate of transferring error free data performance can be measured in terms of Transit time and response time Transit time is the amount of time required for a message to travel from one device to another response time is the elapse time between an inquiry and a response the performance of a network depends on a number of factors including the number of users the type of transmission medium

the capabilities of the connected hardware and the efficiency of the software performance is often evaluated by two networking matrices such as number one throughput throughput is the number of messages successfully delivered per unit time throughput is controlled by available bandwidth as well as the available signal to noise ratio and Hardware limitations the throughput is calculated by dividing the file size by the time to get the throughput in megabits kilobits or bits per second number two delay Network delay is an important design and performance characteristic of a computer network or telecommunications Network the delay of a network specifies how long it takes for a bit of data to travel across the network from one node or endpoint to another it is typically measured in

multiples or fractions of seconds Network delay is divided into several parts such as processing delay processing delay is the time routers take to process the packet header number two queuing delay it is the time the package spends in routing queues transmission delay it is a time it takes to push the packets bits onto the link and the propagation delay propagation delay is a time for a signal to reach its destination we often need more throughputs and less delay however these two criteria are often contradictory if we try to send more data to the network we may increase throughput but we increase the delay because of traffic congestion in the network number two consist consistency consistency is the predictability of response time and

accuracy of data users prefer to have consistent response times they develop a feel for normal operating conditions for example if the normal response time is 3 seconds for printing to a network printer and a response time of over 30 second happens we know that there is a problem in the system accuracy of data determines if the network is reliable if a system loses data then the users will not have confidence in the information and will often not use the system number three reliability the reliability of the network is measured by certain factors such as frequency of failure frequency of failure determines how frequently the network fails recovery time it is the time taken by a device or Network to recover from the

5 minutes, 1 second failure catastrophe Network must be protected from the disasters such as fire and earthquake number four recovery recovery is the Network's ability to return to a prescribed level of operation after a network failure this level is where the amount of lost data is non-existent or at a minimum recovery is based on having backup files number five SC security security of the network is considered as the most important aspect for improving the network performance the network security may be affected due to viruses and unauthorized access of other users to provide network security avoid opening unknown email attachments which may contain virus use anti virus software for securing the systems from

virus firewalls can be implemented for detecting and preventing unauthorized access of other users in the network use backup tools to store the important data on removable media like CD or zip dis this helps to secure your data turn off the system and remove the network cable when not in use to Avid unauthorized interference in the systems now let us learn about physical structure of a network physical structure of a network is described by type of connection and topology of network so what do you mean by type of connection a network is two or more devices connected through links a link is a Communications pathway that transfers data from one device to another for visualization purposes it is

simplest to imagine any link as a line drawn between two points for communication to occur two devices must be connected in some way to the same Link at the same time there are two possible types of connections Point too and multi-point a Point topoint connection provides a dedicated link between two devices the entire capacity of the link is reserved for transmission between these two devices most point topoint connections use an actual length of wire or

cable able to connect the two ends but other options such as microwave or satellite links are also possible when you change television channels by infrared remote control you are establishing a Point to-point connection between the remote control and the television's control

system the examples for Pure Point too links can be two computers communicating via modems a Mainframe terminal communicating with front-end processor or a workstation communicating along a parallel cable to a printer the other type of connection is multipoint connection a multi-point also called multi-drop connection is one in which more than two specific devices share a single Link in a multipoint environment the capacity of the channel is shared either spacially or temporally if several devices can use the same link simultaneously it is a specially shared connection if users must take turns it is a Time shared connection the figure shows both a point to-point and multi-point connections now let us discuss about topology of a network network topology is the

arrangement of the various elements like links nodes etc of a computer network essentially it is the topological structure of a network and may be depicted physically or logically physical topology is the placement of the various components of a network including device location and cable installation the cabling layout used to link devices is a physical topology of the network this refers to the layout of cabling the locations of nodes and the interconnections between the nodes and the cabling the physical topology of a network is determined by the capabilities of the network access devices and media the level of control or fault tolerance desired and the cost associated with cabling or telecommunication

circuits logical topology illustrates how data flows within a network regardless of its physical design distances between nodes physical interconnections transmission rates or signal types May differ between two networks yet their topology may be identical there are four basic topologies possible mesh star bus and ring as shown in the figure let us discuss them one by one number one mesh topology in a mesh Network topology each of the network node computer and other devices are interconnected with one another every node not only sends its own signals but also releases data from other node in fact a true mesh topology is the one where every node is connected to every other node in the network this type of topology is very

expensive as there are many redundant connections thus it is not mostly used in computer networks it is commonly used in wireless networks flooding or routing technique is used in mesh topology mesh topology is shown in the figure and it is of two types number one fully connected mesh topology in fully connected mesh topology every device has a dedicated Point to-point link to every other device the term dedicated means that the link carries traffic only between the two devices it connects to find the number of physical links in a fully connected mesh network with n nodes we first consider that each node must be connected to every other node node one must be connected to n minus one nodes node two must be connected to $n-1$ nodes and finally node n must be connected to n minus 1 nodes so

we need $n-1$ physical links however if each physical link allows a communication in both directions we can divide the number of lengths by two in other words we can say that in a mesh topology we need $n-1$ / two duplex mod links to accommodate that many links every device on the network must have $n-1$ input output ports to be connected to other $n-1$ stations full mesh topology is used only for backbone networks number two partial mesh topology this is far more practical as compared to full mesh topology here some of the systems are connected in similar fashion as in mesh topology while rest of the systems are only connected to one or two devices it can be said that in partial mesh the workstations are indirectly connected to other

devices this one is less costly and also reduces redundancy now coming on to the advantages of mesh topology there are many advantages associated with this topology such as number one data can be transmitted from different devices simultaneously this topology can withstand high traffic number two even if one of the components fails there is always an alternative present so data transfer doesn't get affected number three expansion and modification in topology can be done without disrupting other nodes now disadvantages of mesh topology some of the disadvantages are number one there are high chances of redundancy in many of the network connections number two overall cost of this network is way too high as compared

to other network topologies number three setup and maintenance of this topology is very difficult even administration of the network is difficult then comes star topology in star topology all the components of network are connected to the central device called Hub which may be a hub a router or a switch unlike bus topology where nodes were connected to Central cable where all the workstations are connected to Central device with a Point to point connection so it can be said that every computer is indirectly connected to every other node by the help of the Hub all the data on the star topology passes through the Central device before reaching the intended destination Hub act as a junction to connect different nodes present in Star

Network and at the same time it manages and controls whole of the network depending on which Central device is used Hub can act as a repeater or a signal booster Central device can also communicate with other hubs of different Networks unshielded twisted pair ethernet cable is used to connect 15 meters, 1 second workstations to Central node the central hub also can be classified in the subtypes such as number one passive Hub a passive Hub is a simple signal splitter its main function is to connect the arms of the star while maintaining the proper electrical characteristics a passive hub routes all traffic to all nodes this means that a tremendous load can be created when much communication takes place between computers every computer has the additional burden of reading the address

of each piece of information it receives to determine if the information is intended for that computer information containing addresses of others is discarded number two active Hub an active Hub performs the same function as a passive Hub but contains electronic circuits that regenerate and retransmit the information thus active hubs can be used to extend the size of a

network number three intelligent Hub intelligent hubs perform the same functions as passive and active hubs however they can make informed path selections and perform some Network management intelligent hub root traffic only to the branch of the star on which the receiving node is located if redundant paths exist and intelligent Hub can root information around normally used paths when cable

problems occur routers Bridges and switches are examples of Hub devices that can root Transmissions intelligently intelligent hubs also can incorporate diagnostic features that make it easier to troubleshoot network problems figure shows the star topology now the advantages of star topology number one as compared to bus topology it gives far much better performance signals do not necessarily get transmitted to all the workstations a sent signal reaches the intended destination after passing through no more than three four devices and two to three links performance of the network is dependent on the capacity of the central Hub second Advantage is it is easy to connect new nodes or devices in star topology new nodes can be added easily without affecting rest of the network

similarly components can also be removed easily number three centralized management it helps in the monitoring of the network number four failure of one node or L doesn't affect the rest of the network at the same time it's easy to detect the failure and troubleshoot it next is disadvantages of star topology number one too much dependency on Central device has its own robex if it fails whole network goes down number two the use of Hub a router or a switch as Central device increases the overall cost of the network number three performance and as well as number of the nodes which can be added in such typology is dependent on capacity of the central device now the next apology we are going to learn is the bus topology bus topology is the simplest of

network topologies in this type of topology all the notes bus topology is the simplest of network topologies and this type of topology all the node computers as well as servers are connected to the single cable called bus with the help of the interface connectors this Central cable is the backbone of the network and is known as a bus every workstation communicates with the other device through this bus a signal from the source is broadcasted and it travels to all workstations connected to the bus cable although the message is broadcasted but only the intended recipient whose Mac address or IP address matches accepts it if the Mac or IP address of machine doesn't match with the intended address machine discards the signal a Terminator is added at ends of

the central cable to prevent bouncing of signals a barrel connector can be used to extend it figure shows the bus topology now what are the advantages of linear bus topology number one it is easy to set up and extend bus Network number two cable length required for this topology is the least compared to other networks number three bus topology cost very less and number four linear bus network is mostly used in small networks good for land we also have certain disadvantages of linear first topology first disadvantage is there is a limit on Central cable

length and number of nodes that can be connected number two dependency on Central Cable in this topology has its disadvantages if the main cable

encounters some problem whole network breaks down number three proper termination is required to dump signals use of Terminators is must number four it is difficult to detect and troubleshoot fault at 21 minutes, 1 second individual station number five maintenance cost can get higher with time number six efficiency of bus Network reduces as the number of devices connected to it increases number seven it is not suitable for networks with heavy traffic and number eight security is very low because all the computers receive the sent signal from the source next topology we are going to discuss is ring topology and ring topology all the nodes are connected to each other in such a way that they make a closed loop each workstation is connected to

other two components on either side and it communicates with these two adjacent neighbors data travels around the network in One Direction sending and receiving of data takes place with the help of a token now what is a token and how it is passed token contains a piece of information which along with data is sent by The Source computer this token then passes to the next Node which checks if the signal is intended to it if yes it receives it and passes the empty token into the network otherwise it passes the token along with the data to the next Node this process continues until the signal reaches its intended destination the nodes with tokens are the ones only allowed to send the data other nodes have to wait for an

empty token to reach them this network is usually found in offices schools and small buildings there are many advantages of Ring topology number one this type of network topology is very organized each node gets to the data when it receives an empty token this helps to reduce chances of collision also in ring topology all the traffic Flows In Only One Direction at very high speed second Advantage is even when the load on the network increases its performance is better than that of the bus topology number three there is no need for Network server to control the connectivity between workstations number four additional components do not affect the performance of the network and number five each computer

has equal access to the resources there are also some disadvantages of ring topology such as number one each packet of data must pass through all the computers between source and destination this makes it slower than star topology number two if one workstation or port goes down the entire network gets 24 minutes, 1 second affected number three network is highly dependent on The Wire which connects different components and number four muus and network cars are expensive as compared to the ethernet cars and hubs next is the tree topology tree topology integrates the characteristics of star and bus topology earlier we saw how in physical star Network topology computers are connected with each other through

Central Hub and we also saw in bus topology workstation devices are connected by the common cable called bus after understanding these two Network configurations we can understand tree topology better in tree topology the number of star networks are connected using the bus this main cable seems like a main stem of a tree and other star networks as the branches it is also called expanded star topology ethernet protocol is commonly used in this type of topology the figure shows the tree topology there are many advantages of tree topology number one it is an extension of star and bus topologies so in networks where these policies can't be implemented individually for reasons related to scalability tree topology is the best

alternative number two expansion of network is possible and easy number three the whole network can be divided into segments which can be easily managed and maintained number four error detection and correction is easy number five each segment is provided with the dedicated Point to point wiring to the central Hub and number six if one segment is damaged other segments are not affected tree topology also has some disadvantages such as number one because of its basic structure tree topology relies heavily on the main bus cable if it breaks whole network is crippled number two as more and more nodes and segments are added the maintenance becomes difficult number three scalability of the network depends on the type of the

cable used so with this our today's lecture is over today we have discussed about various Network types and topologies and the advantages and disadvantages associated with them I hope you have understood the concepts in the next lecture we'll discuss about various types of transmission media thank you have a good day